

Speaking with Screens: Design Space and Guidelines for Informational Robot Screens

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Abstract

Advances in AI have enabled robots to engage in flexible, multi-turn dialogue. Yet many scenarios require robots to utilize additional modalities that complement speech to convey complex information. Robot screens can meet this need by supporting parallel processing, quick verification, and richer representations. As robots are integrated into an increasing number of scenarios with complex communication requirements, there is a need to systematically examine how screens may be used and designed to complement and augment verbal communication. In this paper, through an analysis of 357 commercial and research robots, we outline an initial design space of robot screens. Building on this design space, we present findings from two studies: a co-design study ($n = 12$) that explored user preferences for screen designs and derived a set of design guidelines; and an online pilot study ($n = 89$) that integrated these guidelines into screen designs and evaluated how these designs shaped user perceptions of robot communication. Our contributions include a database of screen-equipped robots; the design space of and guidelines for informational robot screens; and an empirical understanding of user perceptions on the use of robot screens.

CCS Concepts

• **Human-centered computing** → **Displays and imagers**; • **Computer systems organization** → **External interfaces for robotics**.

Keywords

Informational robot screens, Design space, Design guidelines

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1 Introduction

Consider the following scenario: the owner of a home instructs a household robot that the living room table must be tidied before guests arrive: “Before my guests arrive this evening, tidy up the living room, but don’t touch the snack tray and board games I left on the

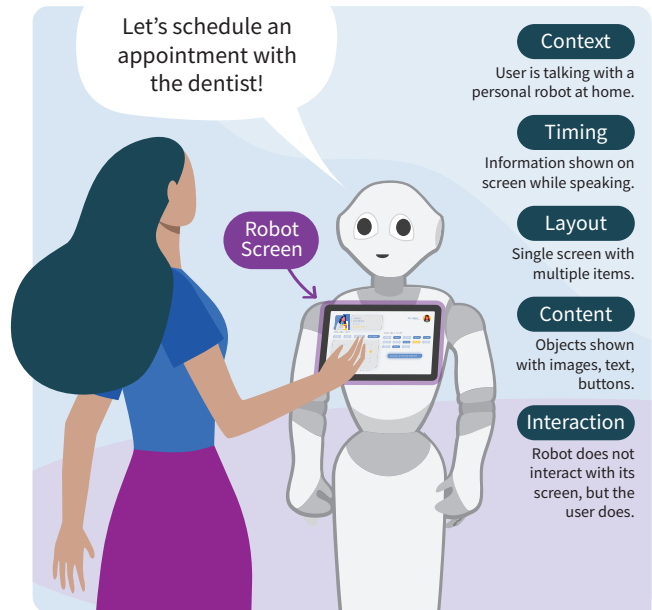


Figure 1: In this paper, we explore the design space and guidelines for informational robot screens and evaluate their effect on user perceptions of robot verbal communication.

coffee table.” To confirm its understanding of the user’s request, on its screen, the robot shows the room cleared, leaving the items mentioned in place, and asks, “Is this correct?” The user quickly verifies the information and confidently expects the outcome that the robot presented. Consider the same scenario, but involving a robot without a screen. The robot would rely solely on spoken communication: “To confirm, I will tidy up the living room but not touch the snack tray and board games you set out for the guests. Is this correct?” In this scenario, the user must pause, listen carefully until the end, and keep track of the details. Such communication can be inefficient, cognitively demanding, and prone to errors.

When robots rely exclusively on speech in their communication, several challenges emerge. First, the sequential nature of spoken language requires listeners to maintain information in working memory until an utterance is complete, increasing cognitive load and the risk of misunderstanding [48]. Second, in time-critical scenarios, the need to wait for the speaker to complete an utterance can delay confirmation and disrupt task flow [28]. Third, certain types of information, such as complex object features or spatial relations, are inherently difficult to express through spoken language



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alone [15, 51]. These limitations persist even as advances in artificial intelligence (AI), particularly large language models (LLMs) enable more flexible, multi-turn dialogue with robots, positioning them as increasingly natural conversation partners [47, 53].

To address these challenges in human-human communication, speakers often employ visual aids such as sketches, notes, and other shared visual references [82, 93]. Such visual support can similarly enhance human-robot communication by enabling richer representations, allowing information to be processed immediately and in parallel, and ensuring overall communication effectiveness. Robot screens, in particular, offer a promising channel for such visual support. Despite this potential, screens remain underutilized, and robots' use of visual aids is mostly limited to facial displays [63]. To address this gap, we introduce the following research questions:

- (1) **RQ1:** What is the design space for robot screens?
- (2) **RQ2:** What are user preferences for elements of this space?
- (3) **RQ3:** How do robot screens affect user experience?

To address these research questions, we first analyzed the screen designs of 357 commercial and research robots to characterize an initial design space for informational robot screens involving several design elements. Through in-person co-design sessions, we conducted an initial exploration of user preferences for these elements, which informed a set of design guidelines. We integrated these elements and guidelines into screen designs and conducted an online pilot study that evaluated how the visual support offered by screens shaped user perceptions of robot communication. The results from the pilot study showed a mixed picture: while the use of a screen improved aspects of usability, such as expectation setting, it decreased other outcomes, such as resilience to failure. Our contributions are as follows:

- (1) Outlining the *design space* for informational robot screens to guide future screen design.
- (2) Creating a *database* of 357 screen-equipped robots to support future investigations of screen design.
- (3) Identifying a set of *design guidelines*, based on human preferences, for screens to serve as active supports for verbal communication.
- (4) Building an *empirical understanding* of how screen designs shape user perceptions of robot communication.

2 Related Works

2.1 Multimodal Support for Robot Dialogue

With recent advances in LLMs, human-robot dialogue has become more common as robots increasingly interact with people in everyday contexts [4, 94]. However, as robots engage in longer and more flexible dialogue, these exchanges often become extended and less predictable. In the settings, when a robot relies on speech, it can be difficult for humans to interpret messages accurately and efficiently. To address this, human-robot interaction (HRI) research has explored supplemental modalities such as gesture and motion. These cues help clarify meaning in dialogue [30, 34], thereby improving human comprehension and communication efficiency [80].

Existing solutions have largely focused on non-visual modalities. In contrast, visual support, particularly a robot's screen, holds strong potential to address these challenges. To date, screen-based

support for verbal communication has been examined mainly in narrow domains, such as presenting educational materials for children [33], providing supplemental captions [68], or visualizing sign language for people with hearing impairments [31]. Yet these applications remain confined to specific contexts. More recently, Ge et al. [27] demonstrated that screen-based visual aids can complement verbal instructions with context, helping users clarify and confirm communication. Existing work implies the broader potential of screens, particularly as LLM-based dialogues enable more multi-turn and open-domain interactions that help users to maintain shared understanding, track context, and resolve ambiguity across turns. However, existing evaluations are often limited to navigation tasks, leaving open questions about their effects in diverse human-robot dialogue contexts.

2.2 Design guidelines for Visuals in HRI

Design principles encompass design considerations, laws, and human biases that guide design work [56]. As Norman [65] emphasized, design guidelines based on broadly applicable principles across technologies are essential for advancing Human-Computer Interaction. In HRI, researchers have similarly explored guidelines for creating usable [36, 54, 92, 96] and human-centered interactions [11, 25, 29, 52, 55, 90] between humans and robots.

Recent studies have examined guidelines for visual modalities, focusing on how robots should present visual information in task and social contexts. For instance, in task guidance, AR-based cues such as overlays or highlights have been studied, with design guidelines for effective cue type in various communication contexts [49]. For social communication, studies have provided guidelines for LED-based gaze displays and reviewed systematic considerations for designing gaze cues to convey emotion and intent [1, 83].

Within this line of research, robot screens have received particular attention in the form of screen-based faces. Prior studies have characterized the design dimensions of robotic faces [43, 98], identified principles for preferred facial features [14, 73], and investigated their influence on human perception [57]. For example, Kalegina et al. [43] analyzed 157 robots with screen-based faces, identifying key facial elements and assessing how they shaped user impressions, ultimately deriving guidelines for face design.

Bridging the Gap. Prior work shows that multimodal support is essential for overcoming the limits of speech-only human-robot dialogue. However, the role of informational screens remains poorly understood and underexplored, with most studies focusing on facial expressions and offering limited knowledge of broader communicative functions and design components. To address this gap, we take an initial step in developing a design space and guidelines for informational robot screens in verbal communication and in investigating the effect of such visual support on user perception.

3 Design Space for Robot Screens

3.1 Robot Screen Data Collection

To identify the design space for robot screens, we first built a database of screen-equipped robots. We searched using the keywords "robot with screen," "robot with display," and "robot with monitor"

Table 1: Design space for informational robot screens.

Design Dimensions	Design Elements	Description and Value Examples
Content	Information Content	The type of information displayed (e.g., emotions [8, 12, 16, 38–41, 44, 50, 59, 67, 85, 86, 95], objects referenced in speech [5, 8, 16, 18, 20, 27, 32, 39, 40, 50, 61, 79, 86, 89]).
	Information Format	The representation format of the information (e.g., text [8, 16, 27, 40, 50, 79, 89], visual objects [5, 12, 16, 18, 27, 32, 38–41, 44, 50, 59, 61, 67, 79, 85, 86, 95]).
Layout	Amount	The total number of screens shown per communication chunk (e.g., one [5, 18, 27, 32, 38, 39, 59, 61, 67, 79, 86, 89, 95], multiple [8, 16, 40, 50]).
	Synchronicity	The number of content types presented simultaneously in a screen within a communication chunk (e.g., one [5, 12, 18, 27, 32, 38, 39, 41, 59, 61, 67, 79, 85, 95], multiple [8, 16, 40, 50]).
Timing	Timing of Information Presentation	The temporal relation of information presentation to the robot’s speech (e.g., before [16, 32, 67], during [5, 8, 12, 16, 18, 27, 32, 38–41, 50, 59, 61, 67, 79, 85, 86, 89, 95], after [16, 32, 39, 50, 67, 89]).
Interactivity	Human–Screen Interaction Modality	The modes through which humans interact with the screen (e.g., touch [8, 50, 79]).
	Robot–Screen Interaction Modality	The modes through which the robot interacts with its own screen (e.g., looking [71], pointing [60, 71]).
	Robot’s Reaction after Interaction	The ways in which the robot react to human–screen interaction (e.g., updating visuals [2, 64], making sound [64], speaking [2]).
Context	Place of Screen Use	The environment in which the screen is used (e.g., home [8, 32, 50, 86], workplace [27, 79, 85, 86, 89], public space [18, 59, 67, 86, 95], hospital [12]).
	Number of Users with the Screen	The number of people interacting with the screen (e.g., one [5, 8, 12, 18, 20, 27, 38–41, 44, 50, 59, 61, 67, 79, 85, 86, 95], multiple [32, 44, 79, 86, 89]).
	Proximity of Screen to Human	The spatial arrangement of the user relative to the screen (e.g., co-located [5, 8, 12, 18, 20, 27, 32, 38–41, 44, 50, 59, 61, 67, 79, 85, 86, 89, 95]).

across general-purpose sources (*i.e.*, Google, YouTube, and commercial platforms such as Amazon). In addition, we used the keywords “Screen(s),” “Display(s),” “Monitor(s)” for robot-related sources (*i.e.*, IEEE robotics databases [35], studies on screen-based robot faces, robot company portfolios, award-winning robots from CES [13]). We also included conceptual robots from the ACM HRI Student Design Competition (2020–2024). Because Google and YouTube searches returned a vast number of results, we stopped our search once no new robots appeared on two consecutive result pages. We excluded robots that used only non-digital LED indicators on black panels (*e.g.*, blinking lights) or required external monitors not attached to the robot. After removing duplicates, the first iteration yielded 368 robots.

In the second iteration, we collected detailed screen-related information from production specifications, user manuals, and demonstration videos, drawing on both official sources and user-generated recordings. This allowed us to code properties such as information type, human interaction method, robot reaction type to screen interactions, number of screens, and screen physical features. For some robots, however, these screen-related details were largely unavailable. Consequently, we removed eleven cases with insufficient screen data, resulting in a final database of 357 robots. The final database is available in Supplementary Materials.¹

¹Supplementary Materials can be found at: https://osf.io/tjeru/?view_only=a9b02eda0b7463ea200be458b2ff996

3.2 Design Space Development

We first grouped the screen-related properties from the 357-robot database into thematic categories (*e.g.*, interaction-related, content-related) to draft candidate design dimensions. In addition to display-related properties, we considered interactivity-related elements, as robot screens are often integrated into the robot’s body and support expressive behaviors such as facial expressions, rather than functioning solely as passive displays. We excluded properties describing fixed physical characteristics (*e.g.*, shape, size, position), as our focus was on the communicative potential of screens rather than relatively inflexible hardware traits.

To situate this draft within human–robot verbal communication, we refined it in two steps. First, we extended the initial dimensions using the design space proposed by Liu et al. [58], which focuses on 2D visual augmentations consistent with most robot screens and emphasizes contextual flexibility. Building on this framework, we added two dimensions *context* and *timing*, which could not be derived from our database due to limited documentation but are critical for verbal communication. Second, we developed value examples for each design dimension based on prior studies in HRI and official robot documentation and guidelines in which screens supported verbal communication, either directly or indirectly.

Finally, we refined the design space in collaboration with two external HRI researchers with ten and six years of HCI research

experience, respectively. This process consolidated overlapping elements and clarified terminology. The finalized design space and value examples are summarized in Table 1.

4 Study 1: Design Guidelines for Robot Screens

To derive actionable, user-driven design guidelines for robot screen design, we conducted a co-design study. Prior work shows that co-design frameworks are effective for generating such guidelines across diverse domains [62, 66, 91].

4.1 Method

4.1.1 Study Design. After providing consent for participation, participants were briefly introduced to the co-design process and then began the session. To scaffold ideation, we provided participants with example images of robots (*i.e.*, Figure 01 [23], Care-O-bot 4 [37], Vita [75], Kiki [46]) equipped with screens in humanoid, machine-like, and zoomorphic forms, representing common categories of robot embodiments [17]. Participants completed co-designs for three human-robot dialogue scenarios (see § 4.1.2), derived from the *Context* dimension of our design space and grounded in existing HRI work.

To accompany the robot’s verbal message in each scenario, participants sketched a robot screen to specify the *Content* dimension and the *Amount* element of the *Layout* dimension. For design aspects not observable visually, namely the *Timing* and the *Interactivity* dimensions, and the *Synchronicity* element of the *Layout* dimension, participants answered co-design questions indicating their chosen values and rationales. Examples were provided for each element, where participants could select multiple options or propose alternatives beyond the design space.

After completing the three scenario-based co-designs, participants took a five-minute break while the research team translated their sketches into visual prototypes. Static designs were presented as static visuals, and animated designs as sequential static frames. These prototypes were then displayed to participants on a Pepper [74] and Double 3 [21] robot. Participants then reviewed if the chosen design options for the *Content* and the *Layout* dimensions are well implemented in the prototypes on both robots, with the robot delivering its verbal message at the participant’s chosen *Timing* dimension. The robot order was counterbalanced across scenarios and participants. This study was approved by the University’s Institutional Review Board (IRB).

4.1.2 Human-robot dialogue Scenarios. We designed three human-robot dialogue scenarios reflecting common interaction contexts.

S₁: Cooking [19, 87]. A human cooking dinner asks a home robot for sugar. Because two types are available, corn sugar and coconut sugar, the robot clarifies: “Do you mean corn sugar?”

S₂: Package Inspection [45, 97]. At a warehouse, a robot informs a human that the ordered ceramic mug arrived at 2:30 PM, adding: “But the box was torn when it was delivered.”

S₃: Museum Tour [26, 69]. During a science museum visit, an educational robot tells children: “The fennec fox has big ears that help it stay cool in the desert!”

4.1.3 Participants. Twelve participants were recruited for the co-design session, and the inclusion criteria required them to be 18

years or older, fluent in English, and have sufficient vision to view a computer monitor. Participants were recruited via university mailing lists. Ages ranged from 18 to 65 years ($M = 32.7$, $SD = 16.9$). Of the twelve participants, five identified as female, six as male, and one as non-binary. Upon completion of the study, participants were compensated at a rate of \$15 per hour.

4.1.4 Analysis. Participants’ screen sketches were analyzed to capture preferences for the *Content* dimension and the *Amount* element of the *Layout* dimension. We inductively coded all sketches to identify common information formats (*e.g.*, images, text, buttons, and icons) and applied this coding scheme across the dataset. Using participants’ accompanying descriptions, we then identified the intended information content (*e.g.*, objects referenced in speech, user-action triggers, emotions) and counted the frequency of each format. Participants’ selections for the *Timing* and *Interactivity* dimensions, and the *Synchronicity* element were also analyzed based on frequency counts using a thematic analysis approach [9]. All sketches are included in the supplementary materials.

4.2 Findings

The co-design study yielded insights across multiple dimensions of our design space. For the *Timing* and the *Interactivity* dimensions, as well as the *Synchronicity* element of the *Layout* dimension, participants showed consistent majority choices and convergent rationales, from which we derived five design guidelines (DGs). In contrast, the *Content* dimension and the *Amount* element of the *Layout* dimension yielded design insights rather than guideline-level generalizations, as participants’ sketches reflected various context-specific considerations.

Each gray box in a subsection indicates the design element targeted by a design guideline or insight. Participants are labeled as P1–P12, where P_i denotes participant i . Figure 2 summarizes participants’ overall preferences for informational screen design.

4.2.1 Synchronicity **DG1: Present related information at once.**

Across all three scenarios, most participants preferred having all information displayed simultaneously rather than sequentially (S_1 : $N = 10$, S_2 : $N = 11$, S_3 : $N = 10$), indicating a consistent preference for synchronous information presentation across contexts. Participants primarily valued this approach for enabling quick and efficient comprehension (S_1 : $N = 6$, S_2 : $N = 11$, S_3 : $N = 7$), which allowed them to continue tasks or make decisions without delay. As one participant explained, “[By seeing all information simultaneously], I wouldn’t spend much time. I read it quickly, and then like boom, I’m done.” (P4). These comments suggest that simultaneous presentation was valued for time-efficient interaction. There was no significant difference between the preference frequencies across all scenarios.

Participants also described how seeing all information at once helped them maintain a coherent understanding without interruption (S_1 : $N = 2$, S_2 : $N = 1$, S_3 : $N = 6$). For example, P8 mentioned “It [Seeing all information at once] just lets people see the whole picture of the question. [...] Getting the whole picture is important so that people have a bigger understanding.” In the S_3 , two participants (P8, P12) highlighted educational benefits, noting that users may

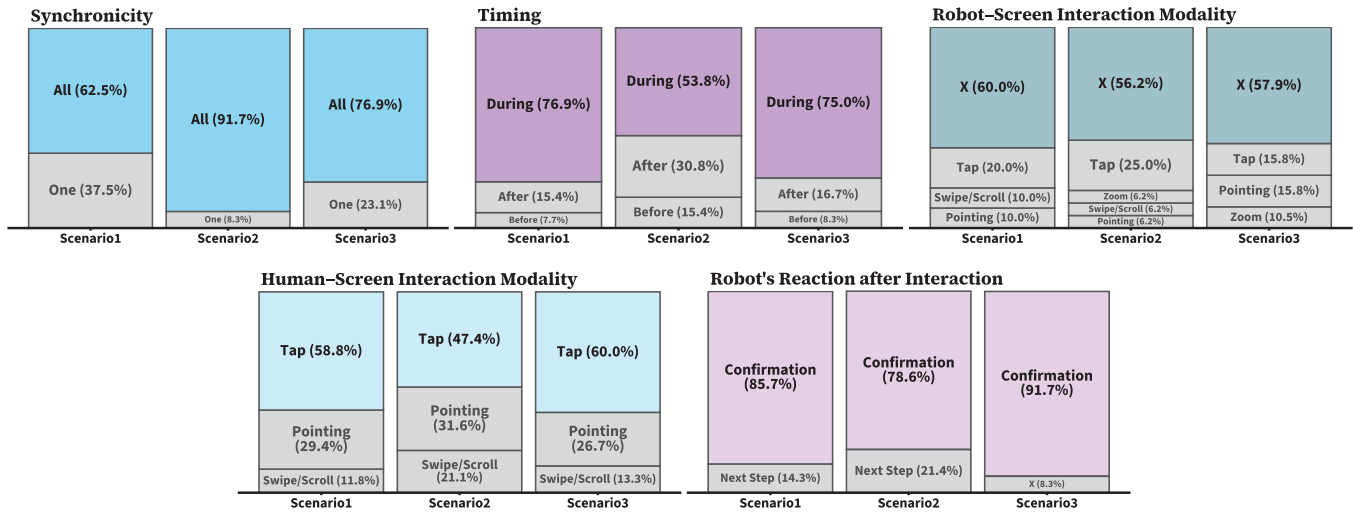


Figure 2: Option preference frequencies for each design element. We note that participants were allowed to select multiple preferred values.

have different interests or learning goals. Simultaneous presentation allowed users to explore information in their preferred order, as P12 described: “Some people just want to know more about the fox, or some people just want to know about like different animals [...] They can choose whatever order they want to go in.”

4.2.2 Timing of Information Presentation DG2: Present information during speech. Across all three scenarios, most participants preferred having visual information presented while the robot was speaking ($S_1: N = 10, S_2: N = 7, S_3: N = 9$). Participants most frequently explained that this timing felt more like a human conversation, in which verbal and visual cues unfold in sync ($S_1: N = 6, S_2: N = 7, S_3: N = 6$). Others noted that synchronized delivery improved efficiency by streamlining interaction, supporting the flow of ongoing activities, and reducing mental load ($S_1: N = 4, S_2: N = 4, S_3: N = 2$). As P9 described, “If it’s before or after, then you’re kind of like temporarily locked out of using the screen or the verbal response.”

Participants’ rationales also varied slightly by scenario. In S_1 , two participants (P1, P9) explained that presenting visuals during speech reduced the risk of missing information when their attention was divided, such as while cooking. As P1 noted, “If it [robot] says, do you mean corn sugar, [...] then you look at it and you would see the screen.” In S_3 , three participants (P3, P11, P12) emphasized the importance of synchronizing visuals with spoken educational content. They pointed out that this approach fits the flow of a lesson (P3), supported different learning styles (e.g., visual learners, verbal learners) by combining modalities (P12), and helped sustain children’s attention by keeping them engaged in real-time (P11).

4.2.3 Human-Screen Interaction Modality DG3: Make the screen tappable. Across all three scenarios, most participants preferred tapping as the primary way to interact with the screen ($S_1: N = 10, S_2: N = 9, S_3: N = 9$). They most frequently described tapping as simple and efficient ($S_1: N = 4, S_2: N = 5, S_3: N = 2$), noting that it

minimized effort and reduced extra steps, allowing them to resolve issues quickly and move on. As P3 stated, “I want to quickly solve the issue. [...] I feel like looking at it and quickly tapping is faster.”

Participants also emphasized that tapping enabled more accurate input ($S_1: N = 2, S_2: N = 2, S_3: N = 4$), as it avoided ambiguity and reduced the risk of misinterpretation compared to gestures such as pointing. In addition, several participants highlighted the familiarity and intuitiveness of tapping ($S_1: N = 3, S_2: N = 1, S_3: N = 1$), noting that it is widely used on other devices and therefore feels natural without requiring additional learning. No meaningful differences were observed in the rationales cited across scenarios.

4.2.4 Robot-Screen Interaction Modality DG4: Avoid robot self-interaction. Across all three scenarios, most participants preferred that robots not interact with the screens ($S_1: N = 12, S_2: N = 9, S_3: N = 11$). Participants commonly viewed such behavior as redundant and unnecessary, noting that the robot’s speech alone was sufficient to convey the information ($S_1: N = 6, S_2: N = 5, S_3: N = 5$). As P11 explained, “I don’t know what it would add by interacting with the screen.”

Participants also expressed concerns that additional robot gestures could be distracting, particularly when they needed to remain focused on primary tasks such as cooking or handling items in a warehouse ($S_1: N = 3, S_2: N = 2, S_3: N = 2$). P3 described this distraction, stating, “I don’t like it because it will distract me from the screen. [...] I’ll try and figure out what the robot is trying to do.” Others noted that these behaviors could slow communication and waste time ($S_1: N = 2, S_2: N = 3, S_3: N = 2$).

Finally, several participants felt that robot self-interaction undermined the naturalness of communication and could feel awkward or inappropriate ($S_1: N = 2, S_2: N = 1, S_3: N = 2$). One participant described the behavior as rude, explaining it could seem to force attention toward the screen: “If it’s like touching or tapping its screen, that’s kind of passive-aggressive. It’s like, did you mean the corn sugar? (Pointing to its screen) So that kind of feels rude.” (P2)

Across scenarios, alternative interaction modalities were rarely preferred. While a small number of participants in S_2 and S_3 mentioned zooming as acceptable, swiping and scrolling were not selected as preferred options in S_3 .

4.2.5 Robot's Reaction after Interaction DG5: Confirm user input. Across all three scenarios, nearly all participants preferred that the robot provide confirmation after users interacted with the screen (S_1 : $N = 12$, S_2 : $N = 11$, S_3 : $N = 11$). Participants most frequently emphasized the importance of clarity and assurance (S_1 : $N = 11$, S_2 : $N = 11$, S_3 : $N = 9$). As P12 explained, “I would like it to give me a response, saying make sure he understand what I’m doing.” Participants described several downstream benefits of confirmation. These included improving efficiency by reducing ambiguity (S_1 : $N = 3$, S_2 : $N = 4$, S_3 : $N = 3$), fostering trust by reassuring users that their input was reliably recognized (S_1 : $N = 5$, S_2 : $N = 2$, S_3 : $N = 2$), and preventing errors by catching misunderstandings before action (S_1 : $N = 2$, S_2 : $N = 3$, S_3 : $N = 1$).

When considering confirmation modality, participants preferred non-visual forms such as brief verbal acknowledgments (e.g., saying “okay”) or nodding (S_1 : $N = 11$, S_2 : $N = 10$, S_3 : $N = 9$). In terms of content, most favored simple acknowledgments that required minimal effort and avoided unnecessary elaboration (S_1 : $N = 7$, S_2 : $N = 9$, S_3 : $N = 5$). Participants also valued affective responses that conveyed friendliness or encouragement, such as praise or supportive feedback (S_1 : $N = 7$, S_2 : $N = 3$, S_3 : $N = 5$). For instance, P6 mentioned, “The most fun thing to do would be for the robot to give a thumbs up.”

Beyond confirmation itself, some participants in S_1 and S_2 expressed interest in the robot indicating what it could do next, suggesting opportunities for follow-up guidance after user interaction.

4.2.6 Information Content, Format, Amount Design insights. In S_1 , participants frequently used images to clarify which type of sugar the user had requested ($N = 9$). They explained that when attention is limited, images support immediate communication and reduce cognitive burden. As P12 mentioned, “I would like to actually have a photo for me [...], Because I think that’s most direct usually when you’re cooking you’re busy, which immediately I can absorb it.” Alongside images, participants often included buttons with symbolic labels to provide clear selectable options ($N = 6$; e.g., Yes/No, ✓/✗). Many designs also paired images and buttons with referent names or short subtitles ($N = 7$), while symbols such as question marks were used to indicate inquiry states ($N = 7$). Participants highlighted that combining visual and textual cues was particularly important in distracting contexts like cooking, where either modality alone might be insufficient. As P5 explained, “With pictures and words, I can always just look and understand it right away.”

In S_2 , images were primarily used to visualize the damaged package, allowing participants to assess the severity of the issue ($N = 10$). As P11 described, “I wanted to show a photo of the damaged package so people can see the problem.” Among those who included images, half also added short supporting text such as arrival times or annotations ($N = 5$). Buttons were commonly used to represent next steps after viewing the image ($N = 8$), while supplemental

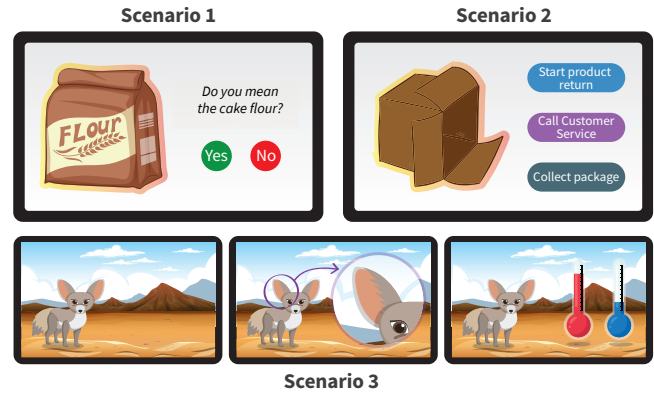


Figure 3: Illustrative screen designs for the three scenarios. In Scenario 3, three sequential screens were designed as an animation.

texts often provided actionable descriptions, such as returning the package or contacting a call center ($N = 10$). Participants repeatedly stressed the importance of explicitly presenting next steps to guide task progression. In addition, three participants (P3–4, 12) proposed distributing the content across two screens, either sequentially or in parallel, to manage information load.

In S_3 , participants widely used images of the fennec fox as supplemental educational material ($N = 10$). They emphasized the importance of dynamic visual content, such as animations, highlighting specific features (e.g., ears), or zooming, as well as displaying temperature trends. These features were seen as essential for maintaining children’s attention. As P4 explained: “I think that change of the image might catch and hold their attention for a while. [...] It would be like a mini cartoon for children.” At the same time, participants expressed concern about overloading the screen with excessive text. Although only a few participants explicitly articulated this during interviews, most avoided long captions and instead used minimal labels (e.g., “fennec fox”) or omitted text entirely ($N = 7$). As P7 remarked: “It would be enough. I don’t think you need much text, because they won’t read it.” Consistent with this concern, seven participants (P2–3, 5–6, 9–11) envisioned splitting content across two sequential screens.

Design insights adopted to screens. The sketch analysis yielded design insights across three elements: information content, format, and amount. For information content, participants commonly incorporated objects referenced in speech to support clarification, efficiency, and understanding across scenarios (S_1 – S_3). For information format, participants frequently used images (S_1 – S_3) and multiple buttons to prompt user actions (S_1 – S_2), often combined with referent names (S_1), supplemental text (S_2), function indicators (S_1 – S_2), or animations (S_3). Regarding information amount, most participants envisioned multiple screens in S_3 , whereas a single screen was preferred in S_1 and S_2 .

Based on these insights, we selected one to two representative screen designs per scenario that captured participants’ dominant preferences. We then refined these designs by adding, removing, or modifying elements that did not align with the consensus. This

process resulted in the final screen designs shown in Figure 3. For S_1 , we replaced the sugar scenario with a flour scenario while keeping other settings unchanged, as corn sugar was not a familiar referent. These finalized designs were used in Study 2.

5 Study 2: Evaluating Robot Screen Visual Support and User Perceptions

We conducted a pilot study evaluating how guideline-informed robot screen-based visual supports influence user perceptions of robot verbal communication.

5.1 Method

5.1.1 Study design. To evaluate the effects of robot screen visual support in human-robot dialogue, we created videos for each of the three scenarios, each lasting approximately 20 seconds. For each scenario, we produced two versions: a screen condition, in which visual information was displayed on the robot's chest screen (Figure 3), and a no-screen condition with no visual information. We conducted a within-subject survey using a Latin square design to control for order effects. Participants viewed one condition per scenario and completed the corresponding questionnaire, resulting in twelve counterbalanced groups.

Videos were created from a first-person perspective with synchronized audio and subtitles to enhance immersion. To isolate the effect of visual support, we applied design guidelines DG1–DG4 derived from Study 1 and excluded DG5, which was outside the scope of this study. After viewing each scenario video, participants completed three questionnaires: a Chatbot Usability Scale we adapted for robots [7], the System Usability Scale (SUS) [42], and the Godspeed Scale [6]. Questions not relevant to robot verbal communication were removed, resulting in a total of 75 questions across all questionnaires.

The survey was conducted using Qualtrics [72]. All frames of the videos and the questionnaires can be found in the supplementary materials. This study was approved by the University's IRB.

5.1.2 Participants. We recruited 89 participants through Prolific [70] who met the inclusion criteria of being 18 years or older, fluent in English, and able to view videos with subtitles. Participants ranged in age from 18 to 65 years ($M = 36.37$, $SD = 12.22$); 43 identified as female and 46 as male. Participants were compensated \$4 for 15 minutes.

5.1.3 Analysis. To analyze the survey data, we first evaluated the reliability of each questionnaire theme [88]. When Cronbach's α was $\geq .70$, questions were aggregated and analyzed at the theme level. When α was $< .70$, we assessed whether factor analysis was appropriate. Specifically, if sampling adequacy was acceptable (Kaiser–Meyer–Olkin $\geq .60$) and eigenvalues exceeded 1, we conducted exploratory factor analysis to extract sub-factors, capturing meaningful latent structures even when overall reliability was modest [22, 81]. If these conditions were not met, analyses were performed at the level of individual questions within each theme.

For the main analysis, we fitted linear mixed-effects models with condition (screen vs. no screen) as a fixed effect. Random effects for participant, scenario, and question were incrementally tested

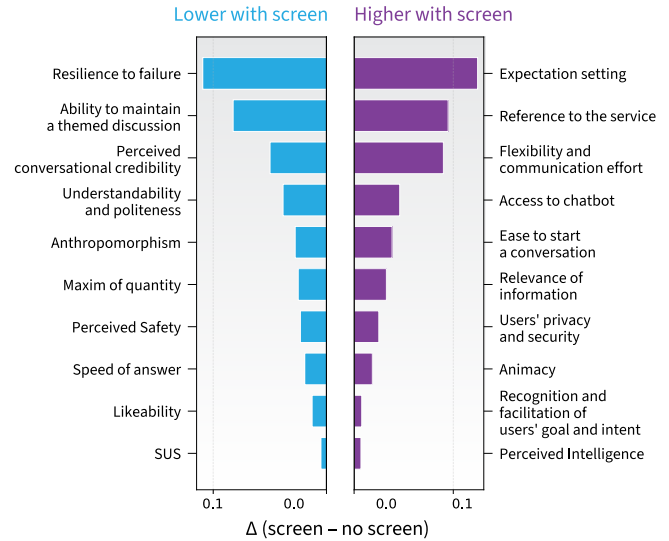


Figure 4: Score differences between screen and no-screen.

and retained only when likelihood ratio tests indicated significant improvement ($p < .05$). Thus, the final random-effects structure varied across themes. When mixed-effects models were unsuitable (e.g., due to negligible variance components), we applied paired-samples t-tests when normality held, or Wilcoxon signed-rank tests otherwise [76, 77]. All pairwise comparisons between conditions were corrected using the false discovery rate. For each analysis unit (theme, sub-factor, or question), we extracted the estimated coefficient representing the difference between conditions (Δ screen–no screen), along with corresponding p-values. All p-values were corrected for multiple comparisons using the false discovery rate, and adjusted results are reported with 95% confidence intervals.

5.2 Findings

Across all measures, mixed-effects models and within-subject tests revealed no statistically significant differences in participants' perception between the two conditions. As shown in Figure 4, the screen condition showed small improvements on several subscales of the Chatbot Usability Scale and the Godspeed Scale (e.g., expectation setting, flexibility and communication effort, animacy), while the no-screen condition showed small improvements on the SUS and other subscales (e.g., resilience to failure, ability to maintain a themed discussion, perceived conversational credibility).

6 Discussion

6.1 Screens as Interactional Support in Dialogue

Our co-design findings suggest that users value robot screens as a means of supporting comprehension and verification during spoken interaction, pointing to a complementary role to expressive or performative uses that are more commonly explored in the literature [e.g., 63, 78, 84]. Across scenarios, participants preferred screens that presented information synchronously with speech, allowing them to process spoken and visual information in parallel. This pattern suggests that screens may enable users to quickly confirm

key details, such as options, constraints, or next steps, without interrupting the conversational flow or requiring users to rely on memory or recall.

In addition, participants consistently favored direct, touch-based interaction with the screen, particularly tapping, over indirect modalities such as pointing. This preference highlights the importance of treating robot screens as interactive artifacts, not merely displays as they were in previous studies [e.g., 3, 10]. Designers should therefore prioritize layouts and affordances that support fast, low-effort input, such as clearly delineated tap targets and minimal navigation depth.

Notably, participants generally preferred minimal robot self-interaction with the screen, expressing discomfort with robots tapping or gesturing at their own displays. This finding suggests that robot–screen interaction may be perceived as redundant or distracting rather than helpful. Instead, users appeared to favor a division of labor in which the robot communicates verbally while the screen provides complementary visual support, reinforcing the robot’s role as an information mediator.

Finally, users consistently emphasized the importance of clear confirmation following screen-based interaction, such as brief verbal or visual acknowledgments that the robot has correctly registered their input. This preference suggests that confirmation serves as a critical mechanism for grounding in screen-supported human-robot dialogue, potentially improving user understanding of how their input will shape the robot’s subsequent behavior and their confidence in the interaction.

6.2 Conditions Shaping Impact of Screens

Our pilot evaluation did not reveal significant differences between screen-supported and speech-only conditions. Rather than concluding that robot screens are ineffective, we highlight important boundary conditions for when screen-based support is likely to influence user perceptions.

We speculate that the lack of differences stems from the low interactional demands of the evaluated scenarios. The tasks were short, single-turn, and not time-critical, requiring minimal information tracking or decision-making. In such contexts, spoken communication alone may already be sufficient, leaving little room for visual support to provide measurable benefits. This explanation suggests that the value of robot screens may become more pronounced in more demanding interactions that may involve higher cognitive load, multi-step reasoning, ambiguity resolution, or sustained engagement over time.

A second potential explanation concerns user attention and the salience of screens as a source of information. Because the screen was an integrated part of the robot, and thus one of the many interactional elements that participants had to attend to, the visual information may sometimes have gone unnoticed. For example, participants may have maintained eye contact with the robot and not have noticed the changes on the screen. This possibility raises a key design challenge: while users expressed a preference for minimal robot self-interaction with the screen, some form of subtle referential cue, such as gaze shifts, head orientation, or deictic language (e.g., “on my screen”), may be necessary to ensure that

visual support is noticed and utilized. Future designs should therefore carefully balance user preferences against overt robot gestures toward the screen with the practical need to guide attention.

These findings also point to methodological implications for evaluating the appropriateness and effectiveness of robot screens. Global perception measures such as SUS alone may be insufficient to capture the benefits of visual support, particularly in low-demand settings. Future evaluations should include task-appropriate outcome measures, such as information recall, error detection, response time, or awareness checks that verify whether users noticed and interpreted screen content. Incorporating such measures would help distinguish between robot screens that are ineffective and screens that are simply underutilized in different contexts.

7 Limitations & Future Work

Our work has several limitations. First, across both studies, our participants experienced generic, decontextualized interaction scenarios, leaving domain-specific communication, such as healthcare, and long-term interactions underexplored, limiting the generalizability of our design guidelines and empirical findings. Second, Study 2 relied on online video-based evaluations, which provide only a limited window into dynamic aspects of human-robot conversational (e.g., how gaze might be integrated, how time-critical information might be differently presented, and how multi-user interactions might be supported in screen designs). Finally, Study 2 did not examine the combined effects of the full set of screen design guidelines. Future lab and field studies comparing screens with and without these guidelines could further validate the contributions of each guideline and the broader applicability of their effects.

8 Conclusion

We investigated how a robot’s informational screen may support human-robot dialogue through the characterization of the design space of screens, identification of design guidelines, and an online pilot evaluation. Across studies, we found new design preferences not yet widely adopted in HRI and nuanced user perceptions of robot’s use of screen-based visual support. Our work provides a structured lens and empirical evidence to guide screen design in HRI and highlights the need to consider screen-based visual supports within the design of embodied robot communication and within broader interaction contexts.

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